EgoCatch: Prehensile Humanoid Catching of Diverse Thrown Objects from Egocentric Depth

Andaman Lekawat[1]

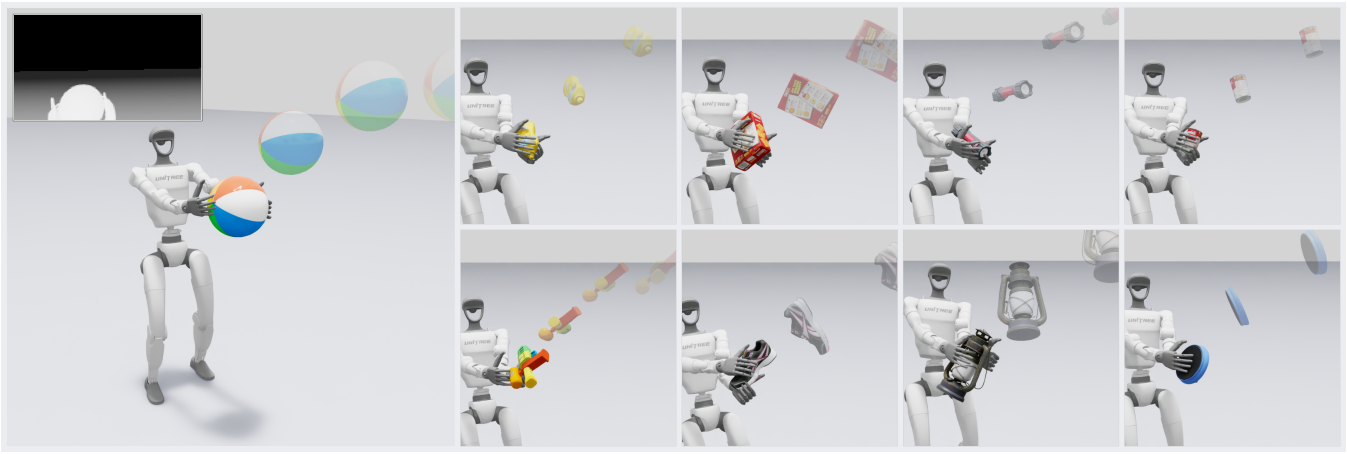


Fig. 1. **EgoCatch performs simulated airborne prehensile catching from egocentric depth and robot state. Left:** interception with synchronized depth; **Right:** Test-512 catches across eight held-out object geometries. Additional rollouts, including failure cases, appear in the supplementary video.

***Abstract*— EgoCatch is a simulated bipedal humanoid that catches and holds unseen thrown objects—512 asset-ID-held-out items from five 3D repositories—using egocentric depth as its only object-specific sensing, with no object pose or external tracking. A privileged controller is trained on a hand-centric distance–velocity (HCDV) channel: at probes on both hands, the closest object surface and the rate at which it approaches. Across matched retrainings, the channel's worst run beats the best run of the privileged object state that prior systems use. Two separate visual systems are derived. EgoCatch-HCDV predicts the channel from depth and keeps the controller frozen; EgoCatch-Action predicts the actions and keeps neither. Five modules per system average $71.0 \pm 5.4\%$ and $72.4 \pm 3.7\%$ under an uninterrupted 2.0 s retention endpoint. Only the first keeps a controller whose input can be exchanged, so we replay identical throws and swap ground-truth HCDV for the visual prediction, an experiment with no hardware equivalent. For the selected system, the swap stays inside a loss margin of 5 percentage points (pp) fixed in advance. The same instrument shows catching improving while pointwise geometric accuracy degrades, and localizes dependence on depth, peaking 150 ms before contact, and on lower-body action.**

I. INTRODUCTION

From a moving, body-mounted view, prehensile catching requires a humanoid to locate a thrown object's surface relative to each hand, estimate its approach, enclose it with articulated fingers, arrest momentum, retain it, and remain balanced. Unlike static grasping, local geometry and approach rate evolve together; unlike blocking or striking, success begins rather than ends at interception.

Humanoids have caught individual ball types [1], [2], [3] and struck them in flight [4], and arm-, hand-, and mobile-manipulator systems broaden prehensile catching across shapes [5], [6], [7], [8], [9]. Together these leave open whether an articulated-finger humanoid can acquire and retain diverse thrown objects from an onboard view, and how a privileged controller should be transferred to it.

Existing approaches leave two coupled gaps. In representation, global object pose and twist leave the finger-local surfaces that determine contact implicit, while static hand–surface geometry omits how quickly those surfaces approach. In transfer, direct visual-to-action learning acquires both implicitly but removes the boundary needed to measure visual substitution independently of control. Hand-centric distance–velocity (HCDV) therefore assigns 22 bilateral probes a closest-surface displacement and signed radial rate.

EgoCatch trains one privileged whole-body controller on ground-truth HCDV (HCDV-GT) and derives two visual systems from it (Fig. 2a). They are alternatives, trained and evaluated separately and never combined. EgoCatch-HCDV, the *channel route*, predicts the controller's object channel from depth and keeps the controller frozen; EgoCatch-Action, the *action route*, outputs the HCDV-trained controller's actions and deploys without HCDV or the controller. In the channel route, supervision and DAgger first ground the output in HCDV geometry; task adaptation against the frozen controller then optimizes catching rather than reconstruction.

Contributions. *(1) System.* EgoCatch is, to our knowledge, the first simulated bipedal humanoid to catch and retain diverse thrown objects in flight with articulated fingers, using egocentric depth as its only object-specific sensing. Five HCDV-supervised visual modules for one Train-selected

[1]Andaman Lekawat is an independent researcher (e-mail: andaman.lekawat@gmail.com).

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controller achieve $71.0 \pm 5.4\%$ across 512 asset-ID-held-out objects under a fixed 2.0 s retention endpoint (Sec. VI-A). *(2) Representation.* Hand-attached surface relations are established; HCDV's addition is an object-inclusive signed radial rate for pre-contact free flight. Under this recipe that rate is what makes the controller trainable: across five matched retrainings it beats three-frame displacement history by 35.4–71.7 percentage points (pp), and one frame without it does not train at all. Across the same retrainings the whole channel also rank-separates completely from the privileged object state that prior systems use (Sec. VI-D). *(3) Instrument.* Retaining the controller behind that channel makes its input auditable: paired replay substitutes HCDV-GT with adapted visual input inside one frozen controller—unavailable to the equal-budget action route. The instrument shows that catching success and pointwise geometric fidelity can move in opposite directions. Two further interventions localize the system's dependencies in time, where depth dependence peaks in the final 150 ms before contact, and in embodiment, where masking lower-body action leaves interception and four-link contact intact while removing securing and retention (Sec. VI-B, VI-C, VI-E).

II. RELATED WORK

Catching systems. Arm-scale systems establish impact-aware capture [10] and privileged-state bimanual catching [7]; free-floating dexterous hands add simulated throw–catch [6]; bimanual systems throw and catch between their own hands [11], [12]; and visual sim-to-real catching uses robot-mounted RGB-D or fixed RGB [8], [9]. Catch & Carry uses head RGB for simulated whole-body ball catch–toss [3]; SMASH demonstrates G1 striking [4]. Simulated humanoids also grasp grounded objects, from privileged object state [13] or egocentric vision [14]. EgoCatch differs in combining free-flight prehensile catching by a bipedal humanoid with egocentric depth as its only object-specific sensing.

Hand-relative geometry. DexRepNet++ includes a dual-hand variant of hand-attached surface relations; RobustDexGrasp uses single-hand closest-surface displacement; and LessMimic distills distance-field cues, including normal and tangential decompositions of link velocity, from egocentric depth [15], [16], [17]. EgoCatch combines probe-to-surface displacement with an object-inclusive signed radial rate in a fixed HCDV channel for pre-contact free flight, used as privileged controller input and as visual supervision; Sec. VI-D tests the explicit rate under one controller architecture and budget.

Privileged-to-visual transfer. Privileged-to-visual transfer includes action imitation [18], onboard-depth humanoid control [19], and visual bridges into frozen manipulation policies [20]. Closest architecturally is rapid motor adaptation, which freezes a privileged policy and regresses its latent input from onboard history under supervision [21]; training the predictor and controller separately is known to degrade closed-loop performance [22]. EgoCatch couples them in the remaining direction, task-adapting the channel against the frozen controller, and transfers at either that channel or the actions. Because existing catching systems differ in embodiment, sensing, objects, and endpoints, we use matched internal controls rather than compare their reported success rates.

III. SYSTEM

We describe the system in build order: robot and task, object channel, privileged controller, visual systems.

A. Robot and task

We simulate a Unitree G1 (29 body DoF) with two five-fingered Inspire FTP hands (12 joints each; 53 articulated joints total) in Isaac Lab/PhysX [23]. Fixed-gain implicit-PD joints run at 200 Hz and the policy at 50 Hz. With the robot standing, each object is launched from 1.4–2.3 m within $\pm 20^\circ$ azimuth toward a target offset by ± 0.10 m, with a 1.35–1.50 m apex, and then flies unforced under gravity. Launch orientation is randomized, and each world-frame component of initial angular velocity is sampled independently from $\mathcal{U}(-1.5, 1.5)$ rad/s. This stable-reach envelope permits reactive steps but excludes pursuit and reacquisition (Sec. VII).

B. HCDV control representation

The controller's only object-specific input is HCDV. Closest-surface displacement locates the nearby object surface relative to each hand probe; the signed radial rate reports whether that surface is approaching or receding and how quickly. Following the closest-surface primitive of RobustDexGrasp [16], 22 probes—one wrist, five knuckles, and five fingertips per hand—query the object's complete visual mesh in the pelvis frame (Fig. 2d). For probe $\mathbf{p}_{i,t}$ and nearest mesh point $\mathbf{s}_{i,t}$, nearest-triangle tie breaking gives the displacements $\mathbf{d}_t$, the rates $\dot{\boldsymbol{\rho}}_t$, and the *object channel* $\mathbf{h}_t$:

$$\begin{aligned} \mathbf{cp}_{i,t} &= \mathbf{s}_{i,t} - \mathbf{p}_{i,t}, & \dot{\rho}_{i,t} &= \frac{d}{dt}\|\mathbf{cp}_{i,t}\|, \\ \mathbf{d}_t &= [\mathbf{cp}_{1,t}^\top, \ldots, \mathbf{cp}_{22,t}^\top]^\top \in \mathbb{R}^{66}, \\ \dot{\boldsymbol{\rho}}_t &= [\dot{\rho}_{1,t}, \ldots, \dot{\rho}_{22,t}]^\top \in \mathbb{R}^{22}, & \mathbf{h}_t &= [\mathbf{d}_t^\top, \dot{\boldsymbol{\rho}}_t^\top]^\top \in \mathbb{R}^{88}. \end{aligned} \tag{1}$$

Here $\dot{\rho}_{i,t}$ is the analytic projection of relative surface velocity onto the probe-to-surface direction; negative values denote approach. Because the relative velocity includes object motion, the rate remains informative when a ballistic surface approaches a nearly stationary probe. HCDV-GT queries the complete visual mesh using simulator state and supplies privileged supervision and the frozen-controller reference. This includes surfaces occluded throughout the egocentric frame stack, so HCDV-GT is not generally reconstructible from vision. Functional substitution instead asks whether partial observation can supply a control-adequate channel.

C. Privileged whole-body controller

Observations and actions. The controller observes $o_t = [x_t^{\mathrm{rob}}, c_t, \bar{a}_{t-1}, \mathbf{h}_t] \in \mathbb{R}^{239}$, where $\mathbf{h}_t$ is the object channel of Eq. (1). Here $x_t^{\mathrm{rob}} = [v_{b,t}, \omega_{b,t}, g_{b,t}, q_t, \dot{q}_t] \in \mathbb{R}^{115}$ is *proprioception*: body-frame base linear and angular velocity, projected gravity, and the positions and velocities of the 53 joints. $\bar{a}_{t-1} \in \mathbb{R}^{33}$ is the previously applied, filtered action. $c_t \in \mathbb{R}^3$ is a planar locomotion-command slot that is fixed to zero in every standing-catch rollout and carries no task information. We call the three non-object inputs

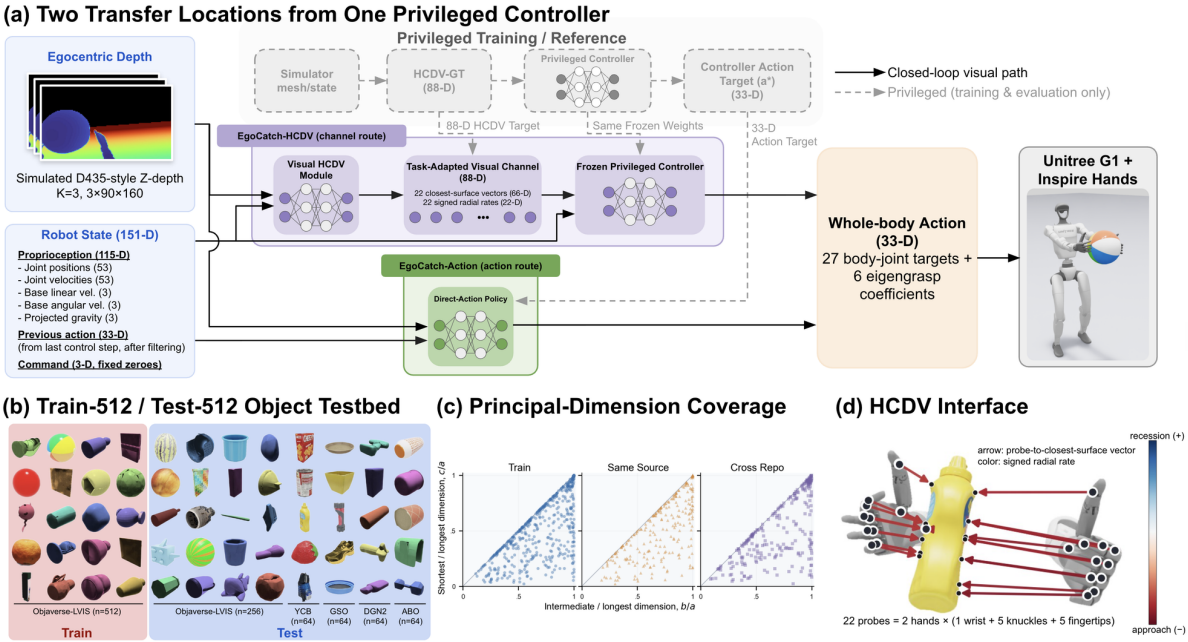


Fig. 2. **Two separate visual systems from one privileged controller. (a)** EgoCatch-HCDV (purple) trains an HCDV predictor, then task-adapts it with the controller frozen; EgoCatch-Action (green) distills actions and deploys without HCDV or the controller. The two are never combined. Paired evaluation swaps only HCDV-GT and the adapted visual channel inside the same controller. **(b)** Train/Test split. **(c)** Principal-dimension coverage per split ($a \geq b \geq c$); same-source denotes held-out Objaverse. **(d)** HCDV probes colored by signed radial rate at an approach-dominated instant.

TABLE I

Controller rewards; penalties negative, contacts force-filtered.

term	wt.	term	wt.
grasp acquisition	$+100$	balance	$+2.4$
secured hold	$+20$	action rate (L_2)	-0.15
grip force (held)	$+10$	forearm contact	-6
fingertip proximity	$+8$	torso contact	-12
uprightness	$+4$	object loss (term.)	-50
close-when-near	$+3$	fall (term.)	-200

TABLE II

Budgets: iterations×envs (task adaptation: iterations×steps×envs).

component	data / budget	key optimization
sphere control	1500×2048	lr 5×10^{-4}
diverse control	1000×2048	lr 10^{-4}; BC 2.0
supervised visual module	1536 eps.; 30 epochs	AdamW 3×10^{-4}
DAgger $\times 2$	$+1536$ eps./round	round-0 standardizer frozen; object-center $z < 0.25$ m excluded
task adaptation	$150 \times 256 \times 32$	actor lr 2.5×10^{-5}

together, $[x_t^{\text{rob}}, c_t, \bar{a}_{t-1}] \in \mathbb{R}^{151}$, the *robot state*. Inputs are clipped to ± 5, and training perturbs base velocity, gravity, and joint position and velocity. The action $a_t \in \mathbb{R}^{33}$ has two parts. Twenty-seven entries are joint position-target offsets (scale 0.25 rad) for the legs, waist, arms, and wrist roll and pitch; the two G1 wrist-yaw joints are not commanded. Six entries are eigengrasp coefficients, three per hand: each hand's three PCA components parameterize its six actuated Inspire coordinates [24]. The eigengrasp coefficients are clamped to ± 1.5. Actions are low-pass filtered before they are applied, $\bar{a}_t = \alpha a_t + (1-\alpha)\bar{a}_{t-1}$, with $\alpha = 0.8/0.5/0.3$ for legs/waist/arm–wrist and 0.1 for eigengrasps.

Reward and termination. Table I gives the reward stack. A one-time acquisition bonus activates after four finger-link contacts above 0.1 N and object speed below 0.5 m/s persist for 0.3 s; dense hold rewards then continue through the 2.0 s retention window inside the 10-s episode. The 0.3-s trigger does not define evaluation success. A no-body-pin gate suppresses acquisition and hold reward on torso or forearm contact; distance shaping uses object-center distance, not HCDV. Evaluation uses the same duration but the stricter predicate of Sec. V.

Training. PPO [25] uses 2048 parallel environments, $[512, 256, 128]$ ELU actor/critic networks, log-parameterized action noise, and zero entropy bonus. Within the stated budget, diverse catching required staged training in our runs: direct 512-shape training did not acquire stable grasps, so a sphere controller warm-starts the diverse stage and anchors it by behavior cloning (coefficient 2.0; Table II).

D. Visual transfer: two separate routes

The two routes are separate systems built from one privileged controller (Fig. 2a), not two heads of one network: each is trained, deployed, and evaluated alone.

Shared input and network. An egocentric D435-style camera supplies 848×480-pixel ($W \times H$) Z-depth over $87.0° \times 56.5°$, pitched $30°$ downward. Values outside 0.20–5.0 m are excluded from validity-aware pooling to 90×160 ($H \times W$); empty bins are zero. During training, depth is corrupted with range-dependent 5–15 mm Gaussian noise, 2% dropout, and 2 mm quantization. Both routes receive three 50-Hz depth frames ($K=3$; 40 ms first-to-last; the first frame repeats at reset) and the robot state of Sec. III-C. Neither receives object pose, twist, box, identity, category, shape code, target, throw, phase, or an external track. Both use identical architectures: a 32/64/64-channel ReLU CNN (kernels 8/4/3, strides 4/2/1), a 256-D projection, and two 512-unit ReLU layers; only the output differs. Data quantities

and optimization budgets are matched (Table II), so the comparison concerns two complete visual-control designs, not one isolated network module.

EgoCatch-HCDV (channel route). The network outputs the object channel, displacement and rate jointly, to the frozen privileged controller. Regression on 1,536 episodes of the privileged controller and two DAgger rounds [18] of 1,536 newly visited episodes each target HCDV-GT and yield the *HCDV predictor*, evaluated for geometric fidelity. *Task adaptation* then runs 150 PPO iterations on the catching reward with the controller frozen and no pointwise HCDV loss, updating only the visual module; we call the adapted module an *adapter* and its output the *task-adapted visual channel*, evaluated for functional substitution. Its layout is HCDV-supervised, but its values are not assumed to reconstruct HCDV-GT; we keep the predictor to measure that transition (Sec. VI-B).

EgoCatch-Action (action route). The network outputs the action directly; neither HCDV nor the privileged controller is present at deployment. Regression and DAgger target the controller's actions, and task adaptation updates the action policy directly.

IV. THE OBJECT TESTBED

Splits and generalization axes. The testbed holds 512 training and 512 test objects from five public 3D sources (Fig. 2b). Train-512 uses LVIS-annotated Objaverse meshes [26], [27]; Test-512 adds 256 asset-ID-disjoint Objaverse instances and 256 cross-repository objects, 64 each from YCB [28], Google Scanned Objects [29], DexGraspNet 2.0 [30], and Amazon Berkeley Objects [31]. A 76-object Test-512 subset exceeds the nominal Train-512 size range. *Unseen* therefore means asset-ID-disjoint from Train-512: the cross-repository half is not internally source-disjoint and categories may overlap, so neither term implies category or geometry disjointness.

Construction without catch outcomes. Train-512 is selected from 794 physically eligible objects by a deterministic, metadata-only rule that never sees catch outcomes: mesh-quality, geometric-catchability, camera-visibility, and 0.35 m/1.5 kg bounds with category-proportional quotas. Test-512 is fixed from identifiers and metadata before evaluation by the same outcome-blind procedure without the size and mass cap. A post-split audit retrieves nearest neighbors by symmetric mean surface distance. Nearest distances are 1.08–5.88% of diameter (median 2.78%), and manual inspection of the closest 32 pairs finds none copied or re-triangulated: coverage, not geometry disjointness (Fig. 2c).

Physical parameterization. Objaverse meshes are deterministically rescaled to category-level real-world size anchors; cross-repository assets retain their source scale. At nominal catalog values, Train-512 spans 0.07–0.35 m in maximum dimension and 0.05–1.50 kg in mass. The 76 out-of-range Test-512 objects extend to 1.0 m in maximum dimension. Nominal masses are *synthetic* category/material priors scaled by volume; training and evaluation randomize them per episode by $\mathcal{U}(0.5, 2.0)$, clamped to 0.3–2.0 kg, with paired banks reusing each sampled mass.

V. EVALUATION PROTOCOL

Catch endpoint. Let $F_{\ell,t}$ be the filtered object-contact force on finger link ℓ, d_t^w the minimum object center-of-mass distance to either wrist, v_t^o object speed, and z_t^o object height. A secure sample is

$$\begin{aligned} C_t &= \sum_{\ell=1}^{24} \mathbf{1}[F_{\ell,t} > 0.1\,\mathrm{N}], \\ S_t &= \mathbf{1}[d_t^w < 0.28\,\mathrm{m} \ \wedge\ v_t^o < 0.20\,\mathrm{m/s} \\ &\qquad \wedge\ z_t^o > 0.60\,\mathrm{m} \ \wedge\ C_t \geq 4], \\ Y_\tau &= \mathbf{1}\left[G_\tau = 1 \ \wedge\ \exists t : \bigwedge_{k=0}^{99} (S_{t+k} = 1)\right]. \end{aligned} \tag{2}$$

Here $G_\tau \in \{0, 1\}$ enforces finger contact before any floor event and rejects torso or forearm contact, falls, and bilateral elbow/wrist overlap from launch through the 2.0 s retention period. Torso contact uses the same force threshold; forearm contact also requires distance < 0.12 m. A fall is root height < 0.30 m or projected-gravity $z > -0.30$; the floor proxy is an oriented-box support point within 0.02 m of ground. Breaks reset the counter, and loss after the period lies outside the primary endpoint. The predicate therefore excludes pickup, bounce, body-pin, shelf, and forearm-wedge strategies while matching the reward's duration; 100 consecutive 50-Hz samples define the stated 2.0 s window (Fig. 3). For failure analysis, we call any sample with $C_t \geq 1$ a *hand contact* and the first sample with $C_t \geq 4$ a *four-link contact*; the latter is not itself a secure catch. We call $\exists t\colon S_t = 1$ *ever-secure* and $Y_\tau = 1$ *retention* for trial τ.

Test scope and selection. Test-512 uses 512 asset-ID-disjoint objects and four throws per object and condition; its 76-object out-of-range subset is extracted from the same rollouts. All trials run to completion. All checkpoints use one schedule and the 2.0 s endpoint on Train-512 for selection. Test-512 is evaluated once after the controller and visual module are frozen, never for training, tuning, or selection; the selected controller is the Train-512 argmax, which scores 71.7% on Test-512, not the Test-512 maximum of 74.7%. Ground-truth object state never enters a visual module's input; it is used for reward shaping, the success predicate, supervision, verification, and diagnostics.

Same-controller functional-substitution test. This test asks whether the task-adapted visual channel can drive the selected HCDV-GT-conditioned controller without an operationally meaningful loss. Paired banks fix object identity, initial robot configuration, pose/twist, and mass; only the object channel $\mathbf{h}_t$ to one frozen controller changes. For each of N objects j in the reported scope and channel source m, four throws give mean success $\bar{y}_j^m$. The reported effect is

$$\bar{y}_j^m = \frac{1}{4}\sum_{e=1}^{4} y_{j,e}^m, \qquad \Delta = \frac{1}{N}\sum_{j=1}^{N} (\bar{y}_j^{\mathrm{vis}} - \bar{y}_j^{\mathrm{GT}}). \tag{3}$$

Unless stated otherwise, every 95% interval uses 10,000 bootstrap resamples of objects, keeping each object's four throws clustered. The same resamples provide marginal

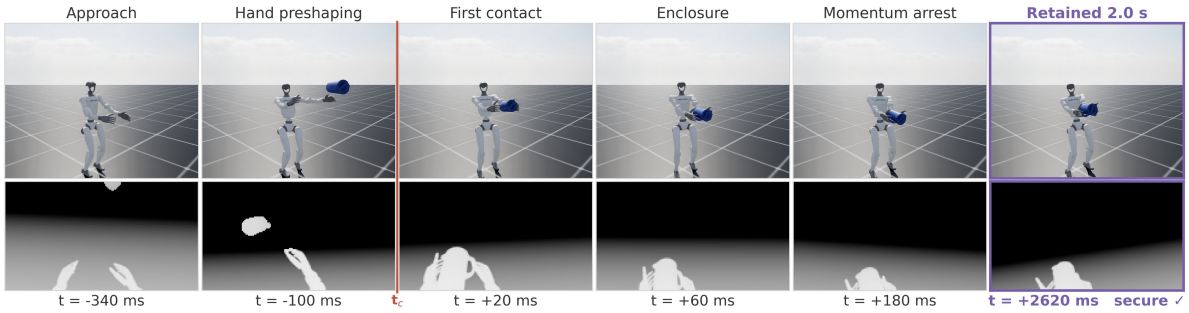


Fig. 3. **One catch, end to end.** The Train-selected EgoCatch-HCDV adapter driving the frozen controller on an irregular non-ball object. **Top:** third-person view, feet and pelvis in frame. **Bottom:** the synchronized egocentric depth the policy receives; times are relative to first contact t_c.

success and paired-effect intervals. Exact matching of initial robot configuration, object state, and throw conditions is available only in simulation. The -5 pp margin was fixed in advance as the largest acceptable aggregate substitution cost: one additional failure per 20 attempts. The endpoint and paired banks were fixed before headline rollouts. Because the experiment asks whether replacing one input channel preserves a fixed controller, bounding possible loss is more informative than testing exact equality.

Estimator and intervention construction. Two hand-built estimators fill the selected frozen controller's object channel from ground-truth-segmented depth, which the visual channel never receives. The *analytic surface* estimator takes each probe's nearest visible surface point for displacement and forms the rate as in Eq. (1), projecting relative velocity, estimated from consecutive depth frames, onto the probe-to-surface direction. The *ballistic tracker* segments unpooled depth, sampled at every second pixel, by keeping the points within 0.75 times the object's largest catalog dimension of its true position. Their centroid, pushed one radius along the view ray, is the observed center of a sphere whose radius is half the object's mean catalog dimension. At every control step a ballistic state under known gravity is fitted to the last two observations; with fewer than four object pixels it is extrapolated. The two-frame window scored highest on 64 Train-512 objects among 2, 4, 8, and 16 frames. Each probe's displacement is the vector to the nearest point on the sphere, with the rate formed in the same way. A sphere has no orientation, so launch spin is not modeled. The budget-matched HCDV-GT control adds a $239 \to 512 \to 512 \to 88$ residual with a zero-initialized head, exactly the identity at initialization, applied after input clipping. The embodiment intervention zeroes the 12 hip, knee, and ankle targets before action filtering. Displacement error uses one fixed 160k-frame bank shared by the HCDV predictor and the adapter, split at 0.30 m probe distance.

Evidence scopes. *Controller retrainings*, five per object input unless marked, test whether that input trains. *Visual modules*, five per route and all derived from the one Train-selected controller, replicate visual transfer, not controller training. *Paired replays* all use that controller, with one Train-selected adapter where vision is in the loop. Within each five-run condition, settings are fixed and runs differ only by seed; $\pm$ is the sample standard deviation (SD) over those runs, and the hand-built estimators are deterministic, with one run each. The route cohorts are independent, and neither is paired with the retrained controllers.

Depth-input diagnostics. Using the selected EgoCatch-HCDV system on full Test-512 (2,048 trials/condition), we shuffle depth across episodes, freeze the first valid stack for the rollout, or delay the live stack by 20 or 160 ms. The interventions alter only depth input; network weights and paired initial conditions remain fixed. We report 2.0 s success. To localize dependence in time, we hold the depth frame for an equal 160-ms window at six onsets relative to t_c, the first contact in the episode's paired unfrozen reference. Each onset uses 256 paired episodes (64 objects $\times$ 4 throws), dose-matched at 2,048 held (env, step) pairs.

Trajectory shifts. Farther/faster throws use 2.5–3.0 m launch distance at the standard apex; high arcs use 1.65–1.90 m apex at the standard distance. Each condition has 512 objects and four method-paired throws per object, and we report each condition as a five-run mean $\pm$ sample SD.

VI. RESULTS

A. Where vision enters: two transfer routes

Both visual routes catch asset-ID-held-out thrown objects using egocentric depth as their only object-specific sensing and retain them for the full 2.0 s (Fig. 1), with five separately trained modules per route sharing one Train-selected controller. EgoCatch-HCDV and EgoCatch-Action reach $71.0 \pm 5.4\%$ and $72.4 \pm 3.7\%$ on standard Test-512 (2,048 trials/run); the Train-selected EgoCatch-HCDV adapter used for all paired tests below scores 72.0% within that cohort. Table III sets both routes beside the main alternative object inputs, all but the no-channel floor receiving privileged information the routes never do.

Route parity. The similar nominal means are consistent with the matched design: both routes share the privileged controller they derive from, inputs, architecture, data, budgets, and the final task-adaptation stage. What differs is auditability: only the channel route keeps a controller whose input can be substituted.

Under farther/faster throws, EgoCatch-HCDV and EgoCatch-Action reach $59.3 \pm 5.4\%$ and $56.9 \pm 3.4\%$; under high arcs, $45.1 \pm 12.5\%$ and $32.7 \pm 8.8\%$. Both conditions lie strictly outside the training launch–apex envelope. The channel route degrades less under both shifts, an ordering we report descriptively because the banks are not

TABLE III

Main methods on Test-512, at the 2.0 s retention endpoint. ±: sample standard deviation (SD) over runs; scopes in Sec. V.

method	controller	success (%)	object information it receives
HCDV-GT, privileged	retrained; 5 runs	68.1 ± 7.7; selected run 71.7	exact HCDV from the complete object mesh
Object State, *the privileged-state input*	retrained; 5 runs	19.1 ± 24.0	object pose and velocity, palm offsets, box size
no object channel	trained; 1 run	0.0	none; robot state only
hand-built analytic surface	selected, frozen; 1 run	18.8	depth + oracle object mask
ballistic tracker, *classical track-and-predict*	selected, frozen; 1 run	60.6	unpooled depth + oracle mask, gravity, size
EgoCatch-Action	none at deployment; 5 modules	72.4 ± 3.7	egocentric depth only
EgoCatch-HCDV	selected, frozen; 5 modules	71.0 ± 5.4	egocentric depth only

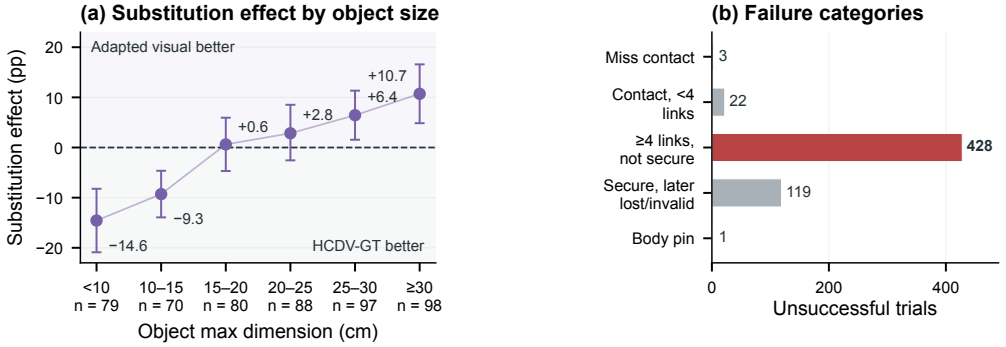


Fig. 4. **Where substitution costs most, and where catching fails. (a)** Paired substitution effect by object size, task-adapted channel relative to HCDV-GT; whiskers are object-clustered 95% intervals, the bins partition one rollout set and their intervals are uncorrected, so the monotone trend rather than any single bin carries the result. **(b)** Of 573 failures among 2,048 trials, most occur after four-link contact.

condition-paired and the five-run spreads overlap.

B. What fills the object channel: privileged, learned, hand-built

Within the channel route, the HCDV predictor—reward-uncoupled in the manner of rapid motor adaptation [21]—reaches 66.1%, a 5.6 pp paired shortfall against the same controller's 71.7% with HCDV-GT. The task-adapted checkpoint reaches **72.0%**, an aggregate increase of 5.9 pp. The task-adapted channel's paired effect is $+0.34$ pp (95% CI $[-2.05, +2.73]$), excluding aggregate degradation larger than the declared -5 pp margin for this selected system. The paired test covers this controller–adapter pair.

Budget-matched control. The visual channel receives 150 adaptation iterations against this controller and HCDV-GT receives none, so the comparison could reflect unequal optimization rather than equivalent information. Granting HCDV-GT the identical budget as an identity-initialized residual yields 74.2% for a paired effect of -2.15 pp: a bounded cost smaller in magnitude than the declared -5 pp margin, and the unmatched HCDV-GT arm reproduces 71.7% unchanged. HCDV-GT is unavailable at test time, so this is an oracle control on optimization effort rather than a deployment configuration. The margin also holds against the unmatched arm when the secure window is extended: measured *unconditionally* on all trials at 3, 5, and 8 s within the 10-s episode, the paired effects are -2.2 $[-4.3, -0.1]$, -2.2 $[-4.6, +0.2]$, and $+0.2$ $[-2.5, +2.9]$ pp, every interval clearing the declared -5 pp margin.

Hand-built estimators. The analytic-surface and ballistic-tracker constructions of Sec. V drive the same frozen controller on the same throws and trail the deployed channel's 72.0% by 53.2 and 11.4 pp despite their privileged inputs (Table III). Both reach hand contact on 97–98% of trials, so the deficit lies after contact rather than at it. The tracker follows the track-and-predict recipe classical ball catching relies on [1], [5]; its shortfall is similar for objects with near-cubic, intermediate, and elongated or flat bounding boxes (-13.6, -10.0, and -11.2 pp), so we find no sign that the sphere approximation drives it.

Geometric grounding and functional adaptation. On 160k held-out frames the supervised predictor has flight/near-contact displacement mean absolute error (MAE) of 6.77/4.61 cm; after controller-frozen task adaptation MAE rises to 7.94/5.06 cm while success rises from 66.1% to 72.0%. HCDV supervision alone thus gives 66.1%, and task reward tunes the channel to this controller for the remaining 5.9 pp, so the deployed output is a task-adapted channel, not a surface reconstruction.

C. How far the visual input can be pushed, and where retention fails

Dependence on aligned depth. Whole-rollout shuffled, frozen, or 160-ms-delayed depth drops 2.0 s success to 0.0–1.2% from a nominal 72.0%; a single 20-ms control step of uniform delay costs 25.8 pp under an adapter trained at zero latency. To ask *when* feedback matters we apply the equal 160-ms freeze of Sec. V at six onsets around first contact t_c, scored on ever-secure, which isolates whether the freeze prevents the grasp from forming. The dependence peaks sharply before contact (Fig. 5c): ever-secure falls 66.0 pp at $t_c - 150$ ms, and 20.3 and 16.4 pp at $t_c - 300$ ms and t_c, all excluding zero, while $+150$, $+300$, and $+600$ ms return intervals containing zero. Depth therefore matters across the approach and peaks in its final 150 ms; onsets strictly after contact leave grasp formation unchanged, so the requirement is concentrated at and before contact.

Visual-input boundary by scale. The substitution effect

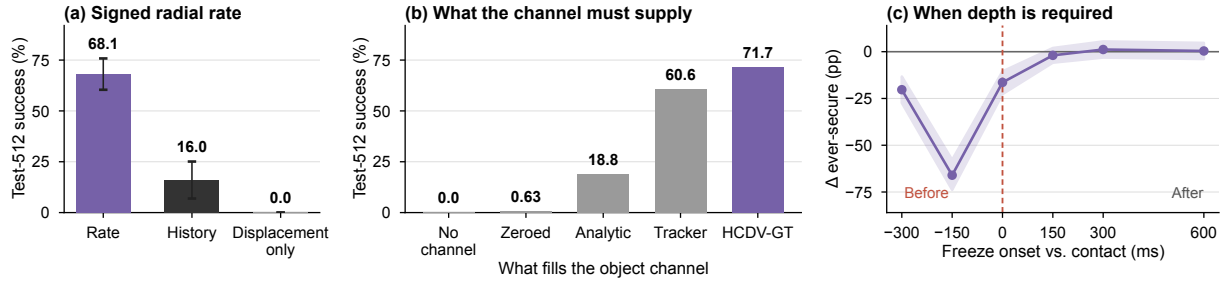


Fig. 5. **Mechanism diagnostics. (a)** Five matched controller retrainings compare rate inputs. **(b)** Test-512 success by what fills the object channel of one frozen controller; the no-channel bar is a separately trained controller, so it bounds the task rather than pairing with the others. **(c)** Paired change in ever-secure under a 160-ms depth freeze, by onset relative to first contact t_c. Error bars: (a) five-run SD; (c) object-clustered 95% band.

rises monotonically with object size: the task-adapted channel trails HCDV-GT by 14.6 and 9.3 pp in the two smallest bins, matches it through the middle, and leads by 6.4 and 10.7 pp in the two largest (Fig. 4a). Small objects occupy few bins in the pooled 90×160 input, the leading candidate cause; the positive tail is consistent with a controller-adapted rather than geometrically faithful channel. Above the Train-512 size cap the substitution still holds on objects reaching 1.0 m—53.0% adapted against 47.4% HCDV-GT on the 76-object subset, a $+5.6$ pp effect with an interval spanning zero—while both conditions fall together, so the limit does not lie in the visual channel.

Capture and retention failures. Of 573 failures, only 25 occur before four-link contact; 428 reach it but never secure, 119 secure before later loss or invalidation, and one is a body pin (Fig. 4b). In this stable-reach envelope, interception is therefore largely solved; converting contact into secure retention dominates failures.

D. Which object representation trains

Signed radial rate. Two ablations remove the rate, replacing the object channel with three stacked frames of its displacement $\mathbf{d}_t$, or one frame of it alone; because input width differs, each repeats the full staged recipe at the same budget, so no arm inherits an HCDV-conditioned initialization. Across five matched controller retrainings, signed radial rate achieves $68.1\pm7.7\%$, versus $16.0\pm9.1\%$ for three-frame displacement history and 0% for one frame without rate (Fig. 5a). Rate wins every matched run by 35.4–71.7 pp against history. Because stacked displacements are in principle rate-sufficient, the gap measures optimization under this feedforward recipe rather than available information.

Task floor. A controller trained with no object channel (robot state only) reaches 0.0% on Test-512, and zeroing the channel on the deployed controller reaches 0.63% (Fig. 5b), against the 71.7% the same controller reaches with HCDV-GT on those throws. The throw envelope alone therefore supports no retained catching; the trained floor is a separate controller, so it bounds the task rather than pairing with the substitution test.

Privileged object input. The 24-D Object State baseline—privileged object state of the kind prior catching systems use [7]—carries the object's heading-frame pose and velocity, its offset to each palm, and its bounding-box size, but no probe-level surface geometry. Under the same staged recipe re-run from scratch, HCDV-GT averages the same $68.1\pm7.7\%$ and Object State $19.1\pm24.0\%$ (Table III), a complete rank separation: the lowest HCDV-GT run, 54.9%, exceeds the highest Object State run, 48.3%. HCDV converges in all five runs; Object State fails outright in two, and its second-best run reaches 42.0%. The separation is therefore one of trainability rather than an information bound under this recipe: HCDV is a task-aligned inductive bias here, not a universal representation ranking. Object State is the standard input of prior catching systems; shape-augmented variants are outside this comparison.

E. How far the result extends, and what the body contributes

Shape and source diagnostics. In the quartile of Test-512 objects farthest from their retrieved Train-512 neighbors, adapted-channel success is 68.9% $[63.3, 74.4]$, an interval containing the 72.0% aggregate, with no detectable monotonic association with shape distance across all 512 test objects (Spearman ρ=0.059, p=0.184), and across sources on the full split it ranges 62.9–85.2%. Success therefore shows no decline with shape distance.

Lower-body dependence. We mask the lower-body action channel at test time on the same frozen controller, in a separate paired intervention replay on a separate throw bank whose own nominal is 71.9% ever-secure and 68.8% retention. The intervention dissociates the phases: hand contact and four-link contact are unchanged (four-link paired Δ $+0.00$ pp, $[-1.95, +1.95]$) while both ever-secure and retention fall to zero. Stance regulation and arrest support are removed together, so the intervention establishes lower-body dependence for the post-contact phase of this controller rather than ruling out an upper-body-only design.

VII. DISCUSSION

What the instrumented route reveals. For a fixed controller, functional input compatibility and pointwise geometric fidelity can diverge, so a visual channel should be judged by closed-loop task performance rather than by reconstruction error. Depth dependence is concentrated at and before contact, so improvements to the visual input should target the pre-contact window.

Simulation and contact. PhysX contact uses CoACD convex decompositions [32] of the same source assets and synthetic inertia; contact is calibrated rather than defaulted, with nine LVIS-derived material classes back-solved against a compliant μ=1.0 fingertip skin to effective pair friction

μ 0.55–1.20 and restitution 0.05–0.45. Objects are rigid, so compliant materials are approximated by damped restitution rather than deformation. HCDV queries the visual mesh, not the collision geometry, so its input is not a collider artifact. Comparisons fix tracking, actuation, and collision geometry and share all physics settings and paired banks, so a perturbation moves both conditions together. Only the sensitivity check varies contact parameters or solver settings: doubling solver position iterations or halving friction changes paired 2.0 s success by -0.4 and -3.5 pp, respectively.

Sensing delay and hardware transfer. Nominal results are at zero sensing latency. The imposed-delay experiment (Sec. VI-C) delays depth alone while the robot state stays current; its 25.8 pp loss for one 20-ms step measures sensitivity to a delay the adapter never saw in training, not end-to-end latency. Compute is not the constraint: pooling, adapter, and controller take 1.0 ms per 20-ms control step on an RTX 4090. Camera exposure, readout, and actuation add hardware-specific delay, so hardware work should measure exposure-to-actuation delay, align depth with robot-state timestamps, and train with randomized delay or prediction. The depth stack uses a simulated corruption model rather than measured hardware noise, and G1–Inspire hardware transfer remains future work.

Scope. The task isolates stable-reach articulated capture; pursuit and reacquisition remain future work. Paired substitution covers one Train-selected controller.

VIII. CONCLUSION

In simulation, EgoCatch catches and retains thrown objects from egocentric depth across 512 asset-ID-held-out objects, at $71.0\pm5.4\%$ and $72.4\pm3.7\%$ for its two routes. Three lessons follow. An explicit approach rate at the hands, rather than a history from which it could be inferred, made the controller trainable. Where vision enters mattered less for success than for what could be verified: both routes caught about as often, but only the channel route kept a controller whose input could be swapped, and for the selected system the swap stayed within a margin fixed in advance while pointwise accuracy fell under task adaptation. Interception is not the bottleneck in this envelope: most failures follow four-link contact, where this controller depends on lower-body action.

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